

VALUATION CHALLENGES OF M&AS

(getting the price range, types of synergies, discounts and premiums)

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Mergers and acquisitions (M&A) are ways companies are combined.

Types of mergers

Mergers happen when:

- Two former independent entities create a new one, while the former ones cease to exist
- Two former independent entities end up being one by one of them being merged into the other

When talking about acquisitions/takeovers, we can differentiate between:

- *Stock acquisition*: the bidder purchases all or part of target's stocks to take control over it
- *Asset acquisition*: the bidder purchases all or part of the target's assets
- *Management Buy-Out*: management already has some stake and purchases all the other, not yet owned, shares
- *Management Buy-In*: getting into the management by a majority stake in the firm, while not owner before

Types of mergers:

- *Horizontal mergers*: firms that are in the same industry, and the same stage of production process; usually buying a competitor (Coca cola + Pepsi)
- *Vertical mergers*: firms that are in the same industry but at different stages of the production process (buying either our supplier or client)
- *Conglomerate mergers*: firms that are different industries (Coca cola + Walt Disney). In this case, since they are two completely different industry, the aim being to create a diversified portfolio to decrease risk.

Focus before the price (Due diligence):

- Verify the financial and business assumptions.
- Assess known liabilities and identify potential liabilities
- Identify potential deal killers

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- Assess the cultural fit (the integration of the two companies, problems can come up post merger)
- Gain enough information to begin preliminary integration planning

Relevant aspects in M&A transactions:

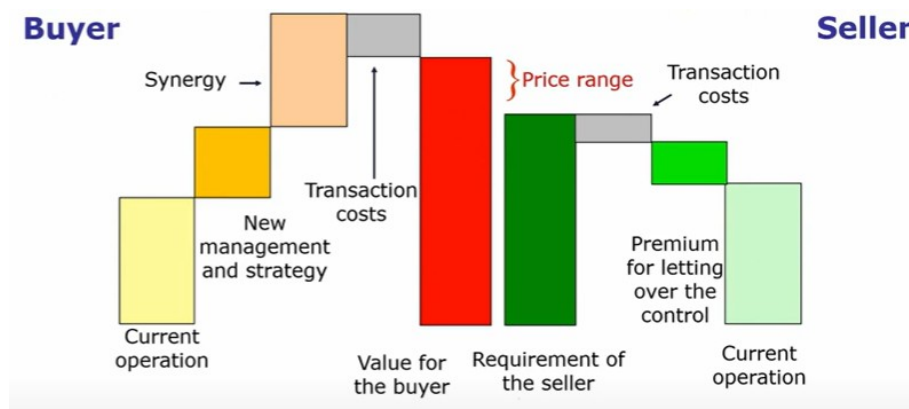
- The analysis of synergies
- The choice of the consideration (the mean of payment)

Price range

When we talk about M&As, there is a bargaining about the buy/sell price, which occurs inside a price range. This price range is between the minimum price the seller would sell the company and the max price the buyer would pay for it.

Why would someone pay more for a firm than its current MV?

- *Synergy*: the combination of the two firms may have higher value than the two separately
- *Control premium*: the ability to better manage the company improving its functioning (current price is based on the current management)
- *Undervaluation premium*: the current price on the market is less than the realistic fair value
- Value for the buyer: to get the max price we can add the values from previous arguments (the total value) – transaction costs (lawyer, consultancy, auditors etc.)
- Value for the seller: current operation + premium for letting to take over the firm (having ideas about modifying current operation) + transaction costs



- The transaction cost is *deducted* from the value for the buyer and *added* to the value for the seller

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- The value estimated for the current operation could be *different* for the two sides
- The **final price** is going to be in the remaining price range: between min for seller and max for buyer

Synergies

Synergies

$$\text{EqV(Combined)} > \text{EqV(Bidder)} + \text{EqV(Target)}$$

Synergies refer to *expected cost savings, growth opportunities, and other benefits that occur as a result of the combination of two companies*. They represent one of the primary value enhancers for M&A transactions, especially when targeting companies in core or related businesses. This notion that “two plus two can equal five” helps support premiums paid and shareholder enthusiasm for a given M&A opportunity. The size and degree of likelihood for realizing potential synergies play an important role in framing the purchase price, and often represent the difference between meeting or falling short of internal investment return thresholds and shareholder expectations. Similarly, in a competitive bidding process, those acquirers who expect to realize substantial synergies can typically afford to pay more than those who lack them.

Due to their critical role in valuation and potential to make or break a deal, buyers need to understand the nature and magnitude of the expected synergies. Successful acquirers typically have strong internal M&A or business development teams who work with company operators to identify and quantify synergy opportunities, as well as craft a feasible integration plan. The buy-side deal team must ensure that these synergies are accurately reflected in the financial model and M&A analysis, as well as in communication to the public markets.

Upon announcement of a material acquisition, public acquirers typically provide the investor community with guidance on expected synergies. Depending on the situation, investors afford varying degrees of credit for these announced synergies, which can be reflected in the acquirer's post-announcement share price. Post-acquisition, appointed company officers are entrusted with garnering the proper resources internally and overseeing successful integration. The successful and timely delivery of expected synergies is extremely important for the acquirer and, in particular, the executive management team. Failure to achieve them can result in share price decline as well as weakened support for future acquisitions from shareholders, creditors, and rating agencies.

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While there has been a mixed degree of success across companies and sectors in terms of the successful realization of synergies, certain patterns have emerged. *Synergies tend to be greater, and the degree of success higher, when acquirers buy targets in the same or closely-related businesses.* In these cases, the likelihood of overlap and redundancy is greater, and acquirers can leverage their intimate knowledge of the business and market dynamics to achieve greater success. In addition, *cost synergies*, which are easily quantifiable (such as headcount reduction and facility consolidation), *tend to have a higher likelihood of success than revenue synergies* and are rewarded accordingly by the market. Other synergies may include tangible financial benefits such as adopting the target's net operating losses (NOLs) for tax purposes, or a lower cost of capital due to the increased size, diversification, and market share of the combined entity.

Synergy is a potential additional value from combining two firms. We differentiate between two different types of synergies:

- **Operational synergies**

Operating synergies are those that allow firms to increase their operating income, increase growth, or do both. They consist of *strategic advantages* (doing more projects, enjoying long-term EVA i.e. higher profit than required, achieving a higher return on investments) and *economies of scale* (increase cost-efficiency and profitability).

- *Revenue synergies*: revenue enhancement due to cross-selling, combination of distribution networks and know-how, *greater pricing power* due to higher market share and reduced competition, etc.
- *Cost synergies*: cost reductions stemming from economies of scales in production and administration, greater purchasing power in the raw material market, improvement in logistics, etc.
- *Asset synergies*: “one-shot” cash flow due to disposal of redundant plants, premises, equipment, etc.

- **Financing synergies**

We count on getting some free cash from the other firm which could be used to finance some positive NPV project; we could get tax reduction; increase in debt taking capabilities (more advantageous conditions).

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- Reduction in the *cost of capital* due to the acquisition of a target with a bad credit rating by a bidder with a good credit rating.
- *Diversification*: The combination of two cash flows streams that are not perfectly correlated (from different businesses or different geographies etc.) reduces the risk.
- *Tax synergies*: transfer of net operating losses from target to bidder (or vice versa) or location of plants or profits in a more tax-efficient part of the world.

Synergies that could not be realized during the first complete business year after the transaction are usually completely lost.

Revenue synergies

Revenue synergies refer to the *enhanced sales growth opportunities presented by the combination of businesses*. A typical revenue synergy is the acquirer's ability to sell the target's products through its own distribution channels without cannibalizing existing acquirer or target sales. For example, an acquirer might seek to leverage its strong retail presence by purchasing a company with an expanded product line but no retail distribution, thereby broadening its product offering through the existing retail channel. Alternatively, a company that sells its core products primarily through large retailers might seek to acquire a target that sells through the professional or contractor channel so as to expand its paths to these markets. An additional revenue synergy occurs when the acquirer leverages the target's technology, geographic presence, or know-how to enhance or expand its existing product or service offering.

Revenue synergies tend to be more speculative than cost synergies. As a result, valuation and M&A analysis typically incorporate conservative assumptions (if any) regarding revenue synergies. Such synergies, however, represent tangible upside that may be factored into the acquirer's ultimate bid price. Investors and lenders also tend to view revenue synergies more skeptically than cost synergies, affording them less credit in their pro forma earnings projections.

Cost synergies

Also referred to as "hard synergies", traditional cost synergies include *headcount reduction, consolidation of overlapping headquarters and facilities, and the ability to buy key inputs at lower prices due to increased purchasing power*. Following the combination of two companies, there is no need for two CEOs, two CFOs, two accounting departments, two marketing departments, or two

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information technology platforms. Similarly, acquirers seek opportunities to close redundant corporate, manufacturing, distribution, and sales facilities in order to trim costs without sacrificing the ability to sustain and grow sales. In some cases, a combination enables the new entity to avoid duplicate CAPEX. For example, the T-Mobil/Sprint merger provided the opportunity to combine forces on building out a costly 5G network rather than each company funding it individually.

Increased size enhances a company's ability to leverage its fixed cost base (e.g., administrative overhead, marketing and advertising expenses, manufacturing and sales facilities, and salesforce) across existing and new products, as well as to obtain better terms from suppliers due to larger volume orders, also known as "purchasing synergies". This provides for economies of scale, which refers to the notion that larger companies are able to produce and sell more units at a lower cost per unit than smaller competitors. Increased size also lends towards economies of scope, which allows for the allocation of common resources across multiple products and geographies. Another common cost synergy is the adoption of "best practices" whereby either the acquirer's or target's systems and processes are implemented globally by the combined company.

Buy side valuation

Valuation analysis is central to framing the acquirer's view on purchase price. The primary methodologies used to value a company—namely, comparable companies, precedent transactions, DCF, and LBO analysis—form the basis for this exercise. These techniques provide different approaches to valuation, with varying degrees of overlap. The results of these analyses are typically displayed on a graphic known as a "football field" for easy comparison and analysis.

A comprehensive buy-side M&A valuation analysis also typically includes an analysis at various prices (AVP) and a contribution analysis (typically used in stock-for-stock deals). AVP, also known as a valuation matrix, displays the implied multiples paid at a range of transaction values and offer prices (for public targets) at set intervals. Contribution analysis examines the financial "contributions" made by the acquirer and target to the pro forma entity prior to any transaction adjustments.

As previously discussed, a "football field" is a commonly used as a visual aid for displaying the valuation ranges derived from the various methodologies. For public companies, the football field also typically includes the target's 52-week trading range, along with a premiums paid range in line

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with precedent transactions in the given sector (e.g., 25% to 40%). The football field may also reference the valuation implied by a range of target prices from equity research reports.

Once completed, the football field is used to help fine-tune the final valuation range, typically by analyzing the overlap of the multiple valuation methodologies, as represented by the bars in the graphic below. As would be expected, certain methodologies receive greater emphasis depending on the situation. This valuation range is then tested and analyzed within the context of merger consequences analysis in order to determine the ultimate bid price.

DCF typically provides the highest valuation, primarily due to the fact that it is based on management projections, which tend to be optimistic, especially in M&A sell-side situations. We have also layered in the present value of \$100 million potential synergies (see bar with diagonal lines in Exhibit 7.9), assuming the acquirer is a strategic buyer. This additional value is calculated by discounting the projected after-tax synergies to the present using the target's WACC (see Exhibit 7.10).

Precedent transactions, which typically include a control premium and/or synergies, tend to follow the DCF in the valuation hierarchy, followed by comparable companies. This hierarchy, however, is subject to market conditions and therefore not universally true. Traditionally, LBO analysis, which serves as a proxy for what a financial sponsor might be willing to pay for the target, has been used to establish a minimum price that a strategic buyer must bid to be competitive. However, while the valuation implied by LBO analysis is constrained by achievable leverage levels and target returns, strong debt markets and other factors may drive a superior LBO analysis valuation.

Discounts & Premiums

After finishing the valuation of a company/BU, we are still not done, since there may be some effects that we were not able to include in the valuation methodology (there is always something that we cannot incorporate into your methodology). In this case we should apply discounts or premiums. These changes has to be applied before contrasting two different valuation methodologies (DCF vs multiple-based) because the use and extent of discounts/premiums could be different across the valuation methods. Also, we may have to use some D/Ps in one valuation method, but different ones in another valuation method.

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In case of DCF method, business outlook and market judgment are reflected in the CF plans and discount rates. For multiples, the industry average reflects the average business outlook of the companies, but if our company is different from average, we should account for that. The market judgement would also be incorporated into multiples. The only valuation method where we don't use D/Ps is the *asset-based method* where we evaluate each asset individually: liquidity issues are already included in the valuation of the individual assets; if we assume liquidation, then we do not account for modification due to ownership structure (in case of liquidation the owners get paid in proportion of their stakes, so even if someone has majority stake, will still get the amount corresponding to her stakes); off-balance sheet items are also included in the valuation. In DCF and multiples we can consider liquidity, ownership structure and off-balance sheet item D/Ps.

Effects	DCF	Multiples	Asset-based
Business outlook	CF plan	Industry average	–
Market judgement	Discount rate	Market ratios	Market prices
Liquidity	Premiums and discounts		Different!
Ownership structure			None!
Off-balance sheet items, others			Should be already included

Before averaging the results of these methods, we should have already applied the corrections, so we do not apply them twice.

Two types of corrections:

- Correction on the value of equity
- Correction on the stake of an owner

Correction on the value of total equity

Liquidity

It can mean that:

- *Assets*: hard to sell
- *Size*: the firm or given stake is hard to sell bc of firm size

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- *Financial power*: the firm is close to bankruptcy

Key person discount

- Some key persons in the company without whom the company could not operate the same way as before (ex. a restaurant with a famous chef, SMEs with “one-man show”)

Complex portfolio without synergy

- A firm owns a wide range of different business units and there is just a small number of top managers who have to deal with all these business units
- If in one business unit there is a disruption in operation, the top management would turn toward that unit and the other units will not be optimally managed

Environmental, legal and dependent liabilities

- There may be a big chance that a new law accepted could destroy the future CF

Long-term liabilities not appearing in the books

- Long-term contracts (may be advantageous or disadvantageous)
- We may be hugely dependent on one of our buyers / suppliers

Correction on the stake of an owner

Liquidity

- Having no buyer for a given stake (ex. for a minority stake in a private firm)
- This discount has nothing to do with the company-level liquidity discount

Control rights

- *Strategic majority*: 75% + 1 vote; you can take all decisions without others agreeing
- *Majority*: 50% + 1 vote; with majority stake you probably get a higher proportion of value; take most of the control of the firm
- *Minority*: <51%
- *Blocking minority*: 25% + 1 vote; the minority without which you could not accept any decision

Additional voting rights

Big package discount

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Success factors of M&A transactions

- *Sound due diligence process*: no skeletons in the closet
- *Stable regulatory and legislative environment*: little changes compared to before M&A expectation
- *Effective integration*: biggest issue
- *Economic certainty*
- *Accurately valuing a target*

Reasons for M&A failures

- *Overpayment*: wrong valuation, competition for takeover etc.
- *Poor management integration*